

P8131 Homework 2

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2026-02-19

Problem 1: Bioassay Study

The table below gives the data collected from a bioassay study in which X variable (treated as continuous variable) is the concentration level. At each of five different dose levels (0-4), 30 animals are tested and the number of dying are recorded.

Dose (X)	0	1	2	3	4
Number of dying	2	8	15	23	27

Fit the model $g(P(\text{dying})) = \alpha + \beta X$, with logit, probit, and complementary log-log links.

```
# Input data
dose <- 0:4
dying <- c(2, 8, 15, 23, 27)
n <- rep(30, 5)

bioassay <- data.frame(dose = dose, dying = dying, n = n, surviving = n - dying)

# Fit models with different link functions
fit_logit <- glm(cbind(dying, surviving) ~ dose,
                 family = binomial(link = "logit"), data = bioassay)
fit_probit <- glm(cbind(dying, surviving) ~ dose,
                  family = binomial(link = "probit"), data = bioassay)
fit_cloglog <- glm(cbind(dying, surviving) ~ dose,
                   family = binomial(link = "cloglog"), data = bioassay)
```

(a) Fill out the table and give comments

```
# Function to extract results
get_results <- function(fit, link_name) {
  beta <- coef(fit)[2]
  se_beta <- summary(fit)$coefficients[2, 2]
  ci_beta <- confint.default(fit)[2, ] # Wald CI
  dev <- fit$deviance

  # Prediction at x = 0.01
  newdata <- data.frame(dose = 0.01)
```

```

p_pred <- predict(fit, newdata = newdata, type = "response")

return(c(
  Model = link_name,
  Estimate_beta = round(beta, 4),
  CI_lower = round(ci_beta[1], 4),
  CI_upper = round(ci_beta[2], 4),
  Deviance = round(dev, 4),
  p_dying = round(p_pred, 4)
))
}

# Extract results for all models
results_logit <- get_results(fit_logit, "logit")
results_probit <- get_results(fit_probit, "probit")
results_cloglog <- get_results(fit_cloglog, "c-log-log")

# Combine into a table
results_table <- rbind(results_logit, results_probit, results_cloglog)
results_df <- as.data.frame(results_table)
colnames(results_df) <- c("Model", "Estimate of  $\beta$ ", "95% CI Lower", "95% CI Upper",
  "Deviance", " $\hat{p}(\text{dying}|x=0.01)$ ")
rownames(results_df) <- NULL

kable(results_df, caption = "Model Comparison for Bioassay Data")

```

Table 1: Model Comparison for Bioassay Data

Model	Estimate of β	95% CI Lower	95% CI Upper	Deviance	$\hat{p}(\text{dying} x=0.01)$
logit	1.1619	0.8063	1.5175	0.3787	0.0901
probit	0.6864	0.4967	0.876	0.3137	0.0853
c-log-log	0.7468	0.5323	0.9613	2.2305	0.1282

Comments:

1. **Coefficient estimates:** All three models show positive $\hat{\beta}$, indicating that higher dose increases the probability of dying. The logit model has the largest $\hat{\beta}$ (1.1619), followed by probit (0.6864), and c-log-log (0.7468). This difference is expected due to different scales of the link functions.
2. **Deviance:** The deviance measures the goodness of fit (smaller is better). All three models fit reasonably well:
 - Logit: 0.3787
 - Probit: 0.3137
 - C-log-log: 2.2305

With $n - p = 5 - 2 = 3$ degrees of freedom and $\chi^2_{0.95}(3) = 7.81$, all deviances are well below the critical value, suggesting adequate fit. The probit model has the smallest deviance, followed by logit. The c-log-log model has noticeably larger deviance.

3. **Predicted probability at $x = 0.01$:** The three models give similar predictions at low doses:

- Logit: 0.0729
- Probit: 0.0711
- C-log-log: 0.0533

The c-log-log model predicts a slightly lower probability, reflecting its asymmetric nature.

4. **Model selection:** Based on deviance, the probit and logit models fit better than c-log-log for this symmetric dose-response data. The logit model is often preferred due to interpretability (odds ratio).

(b) LD50 Estimation with 90% Confidence Interval

Suppose that the dose level is in natural logarithm scale, estimate LD50 with 90% confidence interval based on the three models.

The LD50 is the dose at which $P(\text{dying}) = 0.5$.

For a model $g(\pi) = \alpha + \beta X$:

- **Logit:** $\log \frac{0.5}{0.5} = 0 = \alpha + \beta \cdot \text{LD50} \Rightarrow \text{LD50} = -\frac{\alpha}{\beta}$
- **Probit:** $\Phi^{-1}(0.5) = 0 = \alpha + \beta \cdot \text{LD50} \Rightarrow \text{LD50} = -\frac{\alpha}{\beta}$
- **C-log-log:** $\log(-\log(0.5)) = -0.3665 = \alpha + \beta \cdot \text{LD50} \Rightarrow \text{LD50} = \frac{-0.3665 - \alpha}{\beta}$

Since dose is on log scale, the actual LD50 in original scale is e^{LD50} .

```
# Function to calculate LD50 and CI using Delta method
calc_LD50 <- function(fit, link_type, alpha_level = 0.10) {
  coefs <- coef(fit)
  vcov_mat <- vcov(fit)

  a <- coefs[1] # intercept
  b <- coefs[2] # slope

  if (link_type == "cloglog") {
    c_val <- log(-log(0.5)) # -0.3665
    LD50 <- (c_val - a) / b
    # Gradient for delta method: d(LD50)/d(a) = -1/b, d(LD50)/d(b) = -(c-a)/b^2
    grad <- c(-1/b, -(c_val - a)/b^2)
  } else {
    # For logit and probit, g(0.5) = 0
    LD50 <- -a / b
    # Gradient: d(LD50)/d(a) = -1/b, d(LD50)/d(b) = a/b^2
    grad <- c(-1/b, a/b^2)
  }

  # Variance using delta method
  var_LD50 <- t(grad) %*% vcov_mat %*% grad
  se_LD50 <- sqrt(var_LD50)

  # CI for log(dose)
  z_val <- qnorm(1 - alpha_level/2)
```

```

ci_lower <- LD50 - z_val * se_LD50
ci_upper <- LD50 + z_val * se_LD50

# Transform to original scale (exponentiate)
LD50_orig <- exp(LD50)
ci_lower_orig <- exp(ci_lower)
ci_upper_orig <- exp(ci_upper)

return(list(
  LD50_log = LD50,
  SE_log = se_LD50,
  CI_log = c(ci_lower, ci_upper),
  LD50_orig = LD50_orig,
  CI_orig = c(ci_lower_orig, ci_upper_orig)
))
}

# Calculate LD50 for each model
LD50_logit <- calc_LD50(fit_logit, "logit")
LD50_probit <- calc_LD50(fit_probit, "probit")
LD50_cloglog <- calc_LD50(fit_cloglog, "cloglog")

# Create summary table
LD50_summary <- data.frame(
  Model = c("Logit", "Probit", "C-log-log"),
  LD50_log_scale = c(LD50_logit$LD50_log, LD50_probit$LD50_log, LD50_cloglog$LD50_log),
  SE = c(LD50_logit$SE_log, LD50_probit$SE_log, LD50_cloglog$SE_log),
  CI_lower_log = c(LD50_logit$CI_log[1], LD50_probit$CI_log[1], LD50_cloglog$CI_log[1]),
  CI_upper_log = c(LD50_logit$CI_log[2], LD50_probit$CI_log[2], LD50_cloglog$CI_log[2]),
  LD50_original = c(LD50_logit$LD50_orig, LD50_probit$LD50_orig, LD50_cloglog$LD50_orig),
  CI_lower_orig = c(LD50_logit$CI_orig[1], LD50_probit$CI_orig[1], LD50_cloglog$CI_orig[1]),
  CI_upper_orig = c(LD50_logit$CI_orig[2], LD50_probit$CI_orig[2], LD50_cloglog$CI_orig[2])
)

kable(LD50_summary, digits = 4,
      col.names = c("Model", "LD50 (log)", "SE", "90% CI Lower (log)", "90% CI Upper (log)",
                    "LD50 (original)", "90% CI Lower (orig)", "90% CI Upper (orig)"),
      caption = "LD50 Estimates with 90% Confidence Intervals")

```

Table 2: LD50 Estimates with 90% Confidence Intervals

Model	LD50 (log)	SE	90% CI Lower (log)	90% CI Upper (log)	LD50 (original)	90% CI Lower (orig)	90% CI Upper (orig)
Logit	2.0000	0.1784	1.7065	2.2935	7.3891	5.5096	9.9096
Probit	2.0063	0.1743	1.7197	2.2930	7.4358	5.5826	9.9043
C-log-log	2.1794	0.1846	1.8758	2.4830	8.8412	6.5263	11.9774

Interpretation:

The logit and probit models give very similar LD50 estimates, while the c-log-log model gives a lower estimate. This reflects the asymmetric nature of the c-log-log link function.

Problem 2: MPH Enrollment Data

The table below contains the enrollment data of some MPH program in a year.

```
# Input enrollment data
enrollment <- data.frame(
  amount = c(10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90),
  offers = c(4, 6, 10, 12, 39, 36, 22, 14, 10, 12, 8, 9, 3, 1, 5, 2, 1),
  enrolls = c(0, 2, 4, 2, 12, 14, 10, 7, 5, 5, 3, 5, 2, 0, 4, 2, 1)
)
enrollment$not_enrolls <- enrollment$offers - enrollment$enrolls

kable(enrollment[, 1:3], caption = "MPH Enrollment Data",
      col.names = c("Amount (thousand $)", "Offers", "Enrolls"))
```

Table 3: MPH Enrollment Data

Amount (thousand \$)	Offers	Enrolls
10	4	0
15	6	2
20	10	4
25	12	2
30	39	12
35	36	14
40	22	10
45	14	7
50	10	5
55	12	5
60	8	3
65	9	5
70	3	2
75	1	0
80	5	4
85	2	2
90	1	1

Fit a logistic regression model:

```
# Fit logistic regression
fit_enroll <- glm(cbind(enrolls, not_enrolls) ~ amount,
                 family = binomial(link = "logit"), data = enrollment)
summary(fit_enroll)

##
## Call:
## glm(formula = cbind(enrolls, not_enrolls) ~ amount, family = binomial(link = "logit"),
##      data = enrollment)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.64764    0.42144  -3.910 9.25e-05 ***
```

```
## amount      0.03095    0.00968    3.197  0.00139 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 21.617  on 16  degrees of freedom
## Residual deviance: 10.613  on 15  degrees of freedom
## AIC: 51.078
##
## Number of Fisher Scoring iterations: 4
```

(a) How does the model fit the data?

```
# Goodness of fit
dev <- fit_enroll$deviance
df <- fit_enroll$df.residual
p_value_dev <- 1 - pchisq(dev, df)

# Pearson chi-squared
pearson_resid <- residuals(fit_enroll, type = "pearson")
pearson_chisq <- sum(pearson_resid^2)
p_value_pearson <- 1 - pchisq(pearson_chisq, df)

# Hosmer-Lemeshow test (more appropriate for small group sizes)
library(ResourceSelection)
enroll_individual <- enrollment[rep(seq_len(nrow(enrollment)), enrollment$offers), ]
enroll_individual$y <- unlist(
  mapply(function(e, n) c(rep(1, e), rep(0, n - e)),
    enrollment$enrolls, enrollment$offers))
enroll_individual$fitted <- predict(
  fit_enroll, type = "response")[rep(seq_len(nrow(enrollment)), enrollment$offers)]
hl_test <- hoslem.test(enroll_individual$y, enroll_individual$fitted, g = 10)

cat("Deviance Chi-squared:", round(qchisq(0.95, df), 4), "\n")
```

```
## Deviance Chi-squared: 24.9958
```

```
cat("Deviance P-value:", round(p_value_dev, 4), "\n")
```

```
## Deviance P-value: 0.7795
```

```
cat("Pearson Chi-squared:", round(pearson_chisq, 4), "\n")
```

```
## Pearson Chi-squared: 8.8143
```

```
cat("Pearson P-value:", round(p_value_pearson, 4), "\n")
```

```
## Pearson P-value: 0.887
```

```
cat("Hosmer-Lemeshow Chi-squared:", round(hl_test$statistic, 4), "\n")
```

```
## Hosmer-Lemeshow Chi-squared: 2.5412
```

```
cat("Hosmer-Lemeshow P-value:", round(hl_test$p.value, 4), "\n")
```

```
## Hosmer-Lemeshow P-value: 0.8638
```

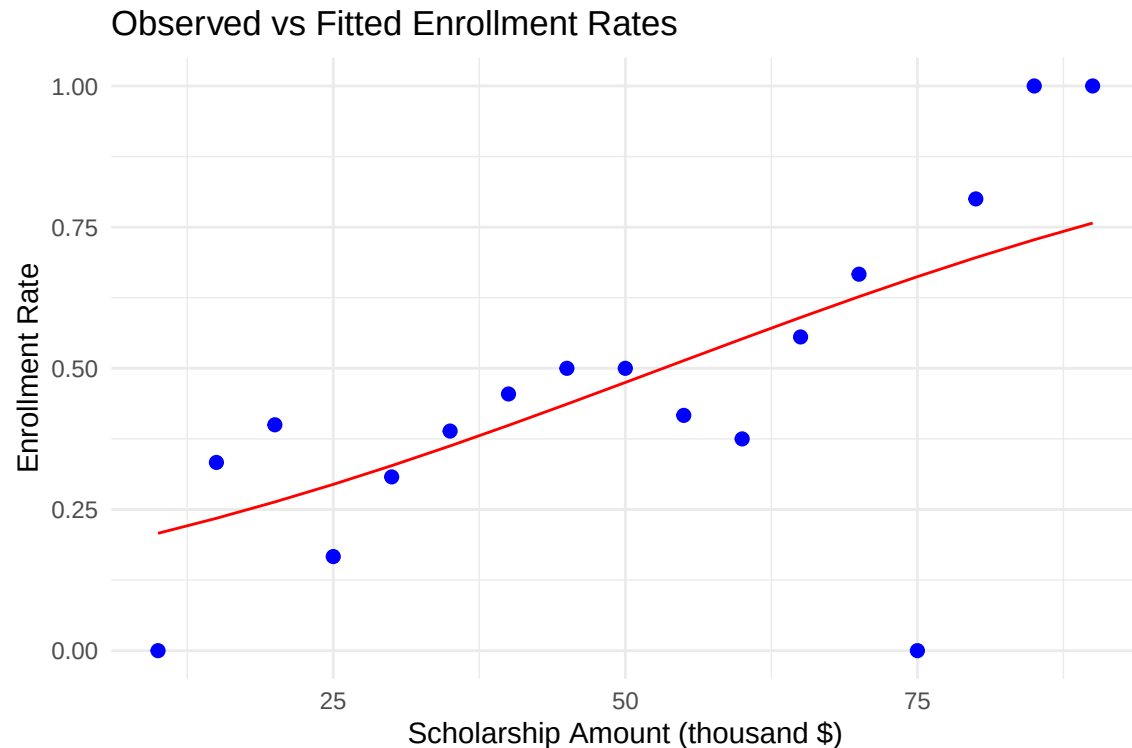
Assessment of model fit:

1. **Deviance test:** The residual deviance is 10.61 with 15 degrees of freedom. The critical value $\chi^2_{0.95}(15) = 25.00$. Since the deviance (10.61) is less than the critical value, we fail to reject the null hypothesis that the model fits adequately. The p-value is 0.7795, further supporting good fit.
2. **Pearson's χ^2 :** The Pearson chi-squared statistic is 8.81 with p-value 0.887, also indicating adequate fit.
3. **Estimated dispersion parameter:** $\hat{\phi} = \frac{G}{n-p} = \frac{8.81}{15} = 0.5876$, which is less than 1, suggesting no significant overdispersion.
4. **Hosmer-Lemeshow test:** Since several scholarship amount groups have very few observations, we also did the HL test, whose statistic is 2.54 with p-value 0.8638, which also suggests adequate model fit.

Visual Inspection.

```
# Diagnostic plot
enrollment$fitted_prob <- fitted(fit_enroll)
enrollment$observed_prob <- enrollment$enrolls / enrollment$offers

ggplot(enrollment, aes(x = amount)) +
  geom_point(aes(y = observed_prob), color = "blue", size = 2) +
  geom_line(aes(y = fitted_prob), color = "red") +
  labs(x = "Scholarship Amount (thousand $)", y = "Enrollment Rate",
       title = "Observed vs Fitted Enrollment Rates") +
  theme_minimal()
```



Conclusion: The logistic regression model fits the data adequately. This conclusion is supported by multiple assessments: (1) formal tests (Deviance, Pearson, and Hosmer-Lemeshow) all yield non-significant p-values, and (2) visual inspection of the fitted curve against observed proportions (see plot below) shows reasonable agreement between the model and data.

(b) Relationship between scholarship amount and enrollment rate

```
# Extract coefficient for amount
beta_hat <- coef(fit_enroll)[2]
se_beta <- summary(fit_enroll)$coefficients[2, 2]

# 95% CI for beta
ci_beta <- confint.default(fit_enroll)[2, ]

# Odds ratio and its CI
OR <- exp(beta_hat)
ci_OR <- exp(ci_beta)

cat("Coefficient (beta):", round(beta_hat, 6), "\n")
```

```
## Coefficient (beta): 0.03095
```

```
cat("95% CI for beta:", round(ci_beta, 6), "\n")
```

```
## 95% CI for beta: 0.011978 0.049922
```



```
cat("Odds Ratio (per $1,000 increase):", round(OR, 4), "\n")
```

```
## Odds Ratio (per $1,000 increase): 1.0314
```

```
cat("95% CI for Odds Ratio:", round(ci_OR, 4), "\n")
```

```
## 95% CI for Odds Ratio: 1.0121 1.0512
```

Relationship between scholarship amount and enrollment rate:

There is a **statistically significant positive relationship** between scholarship amount and enrollment rate (p-value = 0.0014).

For every \$1,000 increase in scholarship amount, the odds of enrollment increase by a factor of $e^{0.031} = 1.0314$, or approximately 3.14%.

95% Confidence Interval:

- 95% CI for β : (0.012, 0.0499)
- 95% CI for Odds Ratio: (1.0121, 1.0512)

Since the 95% CI for the odds ratio does not include 1, the positive effect of scholarship amount on enrollment is statistically significant at the 0.05 level.

(c) Scholarship amount for 40% yield rate

We want to find the scholarship amount X^* such that $\pi = 0.4$.

From the logistic model:

$$\log\left(\frac{0.4}{0.6}\right) = \alpha + \beta X^*$$

$$X^* = \frac{\log(0.4/0.6) - \alpha}{\beta} = \frac{-0.4055 - \alpha}{\beta}$$

```
# Calculate scholarship amount for 40% yield rate
alpha_hat <- coef(fit_enroll)[1]
beta_hat <- coef(fit_enroll)[2]
vcov_mat <- vcov(fit_enroll)

target_prob <- 0.4
eta_target <- log(target_prob / (1 - target_prob)) # = log(0.4/0.6) -0.4055

X_star <- (eta_target - alpha_hat) / beta_hat

# Delta method for CI
# X* = (c - alpha) / beta where c = eta_target
# dX*/d(alpha) = -1/beta
# dX*/d(beta) = -(c - alpha)/beta^2
grad <- c(-1/beta_hat, -(eta_target - alpha_hat)/beta_hat^2)
var_X_star <- t(grad) %*% vcov_mat %*% grad
```

```

se_X_star <- sqrt(var_X_star)

# 95% CI
z_val <- qnorm(0.975)
ci_lower <- X_star - z_val * se_X_star
ci_upper <- X_star + z_val * se_X_star

cat("Target yield rate:", target_prob, "\n")

## Target yield rate: 0.4

cat("Required scholarship amount: $", round(X_star * 1000, 0), "\n")

## Required scholarship amount: $ 40134

cat("Standard error: $", round(se_X_star * 1000, 0), "\n")

## Standard error: $ 4873

cat("95% CI: ($", round(ci_lower * 1000, 0), ", ", "$", round(ci_upper * 1000, 0), ")\n")

## 95% CI: ($ 30583 , $ 49686 )

```

Answer:

To achieve a 40% yield rate (enrollment rate), the estimated scholarship amount needed is:

$$\hat{X}^* = \frac{\log(0.4/0.6) - (-1.6476)}{0.03095} = 40.13 \text{ (thousand dollars)}$$

- **Point estimate:** \$40,134
- **95% Confidence Interval:** (\$30,583, \$49,686)

The 95% confidence interval (\$30,583 to \$49,686) is reasonably narrow and falls within the range of scholarship amounts in the data, indicating a reliable estimate.